

SAT Math Desk Packet

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Goal: Digital SAT Math about 680 → about 780

What this is: Math notes and exam-style questions only. No lesson clock. Use it at the desk: point at a skill, do the items, check the back.

Items are **instructor-created** in Digital SAT style (short stem, 4-choice MC or SPR). They are not copied from Bluebook. Official practice stays in Bluebook / the Student Question Bank.

PSDA = Problem-Solving and Data Analysis (ratios, percents, data, probability, statistical claims). About 15% of Math.

Answer keys are the last section. Do not flip there during a timed set.

1. How Digital SAT Math is built

Fact	Detail
Time	2 modules × 35 minutes, 22 questions each (44 questions, 70 minutes)
Calculator	Allowed on every Math question (Desmos is built into Bluebook)
Formats	About 75% multiple choice (A–D); about 25% student-produced response (SPR)
Adaptive	Module 1 routes Module 2. Easy Module 1 misses can cap a 780
Mix	Every module includes all four domains below

Domain	Share	Typical count	What SAT actually asks
A Algebra	~35%	13–15	Linear equations, linear functions, systems, inequalities
B Advanced Math	~35%	13–15	Equivalent expressions; quadratic / exponential / abs / rational / radical; nonlinear systems
C PSDA	~15%	5–7	Ratios, rates, units, percents, tables, scatterplots, probability, margin of error, study claims
D Geometry & Trig	~15%	5–7	Area and volume, similar triangles, Pythagorean, special rights, sin cos tan, circles

SPR: type a number (integer, decimal, or fraction). Blank SPR is a free miss.

2. Math notes (what to have in hand)

Algebra

- Isolate: expand $\rightarrow$ combine like terms $\rightarrow$ divide. If you see $5 - 2(3x + 1)$, the minus hits **both** terms inside.
- Fractions: multiply **every** term by the LCD.
- Slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$. Parallel = same m . Perpendicular = **negative reciprocal**, so $m_1 m_2 = -1$.
- $y = mx + b$: m is the rate (units of y per 1 unit of x); b is the value when $x = 0$.
- System: 1 solution (intersect), 0 (parallel distinct), infinitely many (same line).
- Inequality: flip **only** when multiplying or dividing by a negative.

Advanced Math

- “Equivalent” = same value for all allowed x . Factor a common binomial, or expand, or graph both.
- Quadratic $ax^2 + bx + c$: zeros from factors; vertex $x = -\frac{b}{2a}$; vertex form $a(x - h)^2 + k$ has vertex (h, k) .
- If $a > 0$, minimum is k . If $a < 0$, maximum is k .
- Exponential: $y = A \cdot r^x$. $A = y(0)$. Growth $r > 1$; decay $0 < r < 1$. “+20%” means $\times 1.20$, not +20.
- Period: $A(1 + r)^{kt}$ completes k growth cycles per 1 unit of t . Months per cycle = $12/k$ if t is in years.
- $|A| = k$: if $k < 0$, no solution; if $k = 0$, one; if $k > 0$, two cases.
- Rational: multiply by LCD; throw away any x that makes a denominator 0.
- Radical: isolate, square, **check** in the original (extraneous roots are a real SAT skill).

PSDA

- Ratio $a : b$ with first part A : $\frac{a}{b} = \frac{A}{B}$.
- Combined rate: add the rates, then time = $\frac{\text{work}}{\text{combined rate}}$.

- Percent of: part = $p \times$ whole. Percent change = $\frac{\text{new} - \text{old}}{\text{old}}$.
- Successive percents multiply: up 10% then up 10% is $\times 1.1 \times 1.1 = 1.21$, not +20%.
- “Increase x by $p\%$ ” means $x(1 + \frac{p}{100})$. Increase by 400% is $\times 5$, not $\times 4$.
- “30% of the 40% that are red” is $0.3 \times 0.4 = 0.12$, not $40 - 30$.
- Mean follows outliers; median often does not. Mean of n values $\Rightarrow$ sum = $n \cdot \text{mean}$.
- Two-way table: $P(A)$ uses the grand total; $P(A \mid B)$ first restrict to the B row or column.
- Margin of error: $52\% \pm 4\%$ means the true value is **likely between 48% and 56%**, not “exactly 52%.”

Geometry and trigonometry

- Linear scale $k \Rightarrow$ area $\times k^2$, volume $\times k^3$.
- Similar figures: sides and perimeter $\times k$; **angles stay the same**.
- Triangle angles sum to 180° .
- Pythagorean: $a^2 + b^2 = c^2$. Families: 3-4-5, 5-12-13, 6-8-10, 9-12-15.
- 45-45-90: legs s , hypotenuse $s\sqrt{2}$.
- 30-60-90: short s (opp. 30°), long $s\sqrt{3}$, hypotenuse $2s$.
- $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\cos \theta = \frac{\text{adj}}{\text{hyp}}$, $\tan \theta = \frac{\text{opp}}{\text{adj}}$. Also $\cos \theta = \sin(90^\circ - \theta)$.
- Circle: $C = 2\pi r = \pi d$, $A = \pi r^2$, arc = $\frac{\theta}{360} \cdot 2\pi r$, sector = $\frac{\theta}{360} \cdot \pi r^2$.
- $(x - h)^2 + (y - k)^2 = r^2$: center (h, k) , radius r .
- Tangent at P : the radius to P is **perpendicular** to the tangent. Find m_{radius} , then $m_{\text{tangent}} = -\frac{1}{m_{\text{radius}}}$, then point-slope **at** P (not at the center).

Combo moves (this is what SAT actually asks)

Isolated formulas above are not enough. These collisions are current leaks — see `../error-log/NOTEBOOK.md`.

- Successive percents **multiply**: up 10% then up 10% is $\times 1.1 \times 1.1 = 1.21$, not +20%. Up then down the same percent does **not** return to the start.
 - “Increase x by 400%” means $5x$, not $4x$.
 - Circle + tangent = geometry **plus** linear equations: radius slope, $m_1 m_2 = -1$, then a line.
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3. Algebra questions

Short stems, SAT phrasing. MC unless marked SPR. Desmos allowed.

A1. The functions f and g are defined by $f(x) = x + 7$ and $g(x) = 7x$. What is the value of $4f(2) - g(2)$?

- A) -5 B) 9 C) 22 D) 36

A2. The graph of $5x + 2y = 10$ in the xy -plane is a line. What is the slope of the line?

- A) $-\frac{5}{2}$ B) $-\frac{2}{5}$ C) $\frac{2}{5}$ D) $\frac{5}{2}$

A3. The cost C , in dollars, of an order of n shirts is $C = 12n + 45$. Which statement is the best interpretation of the number 12 in this equation?

- A) The cost of 12 shirts
B) The cost, in dollars, of each additional shirt
C) The fixed fee, in dollars
D) The number of shirts that cost \$45

A4. A gym charges a \$40 signup fee plus \$25 each month. After m months the total paid is \$165. Which equation represents this relationship?

- A) $40m + 25 = 165$ B) $40 + 25m = 165$ C) $40 + 165m = 25$ D) $25 + 165m = 40$

A5. What is the solution to the system $x + y = 10$ and $2x - y = 5$?

- A) $(5, 5)$ B) $(4, 6)$ C) $(6, 4)$ D) $(3, 7)$

A6. (SPR) Adult tickets cost \$12 and child tickets cost \$7. A group buys 15 tickets for \$145. How many adult tickets did they buy?

A7. How many solutions does the system $2x - 4y = 6$ and $x - 2y = 5$ have?

- A) Zero B) Exactly one C) Exactly two D) Infinitely many

A8. Which values of x satisfy $-2x + 5 > 13$?

- A) $x > -4$ B) $x < -4$ C) $x > 4$ D) $x < 4$

A9. Which point (x, y) is a solution to both $y \leq -x + 2$ and $y \geq 0$?

- A) $(3, 0)$ B) $(0, 3)$ C) $(1, 0.5)$ D) $(-1, 4)$

A10. A line in the xy -plane passes through $(2, 5)$ and $(6, 13)$. Which equation represents the line?

- A) $y = 2x + 1$ B) $y = 2x - 1$ C) $y = \frac{1}{2}x + 4$ D) $y = 8x - 11$

A11. Which equation represents the line that is perpendicular to $y = 3x - 1$ and passes through $(0, 4)$?

A) $y = 3x + 4$ B) $y = -\frac{1}{3}x + 4$ C) $y = \frac{1}{3}x + 4$ D) $y = -3x + 4$

A12. (SPR) $\frac{5x - 2}{4} = \frac{x + 6}{2}$. What is the value of x ?

A13. (SPR) A tank contains $W = 120 - 4t$ liters of water after t minutes. After how many minutes is the tank empty?

A14. For which values of k is $x = 2$ a solution of $3x + k > 10$?

A) $k > 4$ B) $k < 4$ C) $k > 16$ D) $k = 4$

A15. What is the solution of $7 - 3(2x - 1) = 2x + 8$?

A) $x = \frac{1}{4}$ B) $x = 0$ C) $x = 1$ D) $x = 2$

A16. (SPR) A phone plan costs \\$35 per month plus \\$0.10 per text. The total bill is \\$47. How many texts were sent?

4. Advanced Math questions

B1. Which expression is equivalent to $2x(x + 4) - (x + 4)$?

- A) $(x + 4)(2x - 1)$ B) $(x + 4)(2x + 1)$ C) $2x^2 + 7x - 4$ D) $2x^2 + 8x - 4$

B2. Which expression is equivalent to $\frac{x^2 - 9}{x - 3}$ for $x \neq 3$?

- A) $x - 3$ B) $x + 3$ C) $x^2 - 3$ D) $9 - x$

B3. (SPR) $(3x^2y)^2 = ax^by^c$, where a , b , and c are positive. What is the value of $a + b + c$?

B4. Which expression is equivalent to $8x^3 - 27$?

- A) $(2x - 3)^3$ B) $(2x - 3)(4x^2 + 6x + 9)$ C) $(2x - 3)(4x^2 - 6x + 9)$ D) $(8x - 27)(x^2 + 1)$

B5. The function f is defined by $f(x) = 2(x - 3)(x + 1)$. What are the zeros of f ?

- A) 3 and -1 B) -3 and 1 C) 2 and 0 D) 6 and -2

B6. The function g is defined by $g(x) = x^2 + 55$. What is the minimum value of $g(x)$?

- A) 3025 B) 110 C) 55 D) 0

B7. (SPR) For $y = x^2 - 8x + 10$, what is the x -coordinate of the vertex of the graph in the xy -plane?

B8. (SPR) What is the maximum value of $y = -(x + 2)^2 + 9$?

B9. What are the solutions of $|3x + 1| = 10$?

- A) 3 and -3 B) 3 and $-\frac{11}{3}$ C) $\frac{11}{3}$ and -3 D) 3 only

B10. How many distinct real solutions does $|x - 2| = -5$ have?

- A) Zero B) One C) Two D) Infinitely many

B11. (SPR) What is the solution of $\frac{x + 3}{x - 1} = 2$?

B12. Which is a valid solution of $\sqrt{2x + 3} = x$?

- A) -1 only B) 3 only C) -1 and 3 D) There are no valid solutions

B13. What is the solution of $2^{x+1} = 16$?

- A) $x = 2$ B) $x = 3$ C) $x = 4$ D) $x = 8$

B14. (SPR) A population starts at 200 and grows by 50% each decade. What is the population after 2 decades?

B15. In the xy -plane, how many intersection points do $y = x^2$ and $y = 2x + 3$ have?

- A) 0 B) 1 C) 2 D) 3

B16. (SPR) The function f is defined by $f(x) = 3 \cdot 2^x$. What is $f(4)$?

5. PSDA questions

- C1.** (SPR) A mix uses flour to sugar in the ratio 2 : 5. If 12 cups of flour are used, how many cups of sugar are needed?
- C2.** (SPR) A car travels 180 miles in 3 hours. At that constant rate, how many miles does it travel in 2.5 hours?
- C3.** In a group of items, 40% are red. Of the red items, 30% also have stripes. What percent of all the items are red and have stripes?
A) 10% B) 12% C) 70% D) 75%
- C4.** A price of \\$80 increases by 25% and then the new price decreases by 25%. What is the final price?
A) \\$80 B) \\$75 C) \\$85 D) \\$70
- C5.** (SPR) 18 is 45% of what number?
- C6.** A population of 800 grows by 10%, then grows by 10% again. What is the new population?
A) 960 B) 968 C) 880 D) 1000
- C7.** (SPR) What number, increased by 20%, equals 72?
- C8.** For the list 2, 2, 3, 7, 16, what is the median?
A) 2 B) 3 C) 6 D) 7
- C9.** (SPR) Karina's mean on five tests is 80. Four of the scores are 72, 88, 90, and 70. What is the fifth score?
- C10.** Club members:

	Yes	No	Total
Junior	12	8	20
Senior	18	12	30
Total	30	20	50

If a member is selected at random from the **seniors**, what is the probability the member said Yes?

- A) $\frac{18}{50}$ B) $\frac{18}{30}$ C) $\frac{30}{50}$ D) $\frac{18}{20}$

C11. A random sample of voters shows 52% support, with a margin of error of 4%. Which conclusion is most appropriate?

- A) Exactly 52% of all voters support the candidate.
- B) The true support is likely between 48% and 56%.
- C) The candidate will win.
- D) The sample is biased because the result is not 50–50.

C12. A researcher asks only students in the honors hallway whether they like a new schedule, then claims the **whole school** supports it. The main problem with this claim is:

- A) The mean is undefined.
- B) The sample is not representative of the school.
- C) The margin of error must be 0.
- D) This is a well-designed experiment.

C13. The density of a wood is 8 grams per cubic centimeter. A sample has mass 40 grams. What is the volume of the sample, in cubic centimeters?

- A) 5 B) 32 C) 48 D) 320

C14. Using the club table in C10, what is $P(\text{Senior and Yes})$?

- A) $\frac{18}{30}$ B) $\frac{18}{50}$ C) $\frac{30}{50}$ D) $\frac{18}{20}$
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6. Geometry and trigonometry questions

D1. (SPR) What is the volume of a cube with edge length 3?

D2. Every edge of a cube is multiplied by $\frac{1}{2}$. The new volume is what fraction of the original volume?

- A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{1}{8}$ D) $\frac{1}{16}$

D3. Two similar paintings have corresponding side lengths in the ratio 2 : 3. The smaller painting has area 20. What is the area of the larger painting?

- A) 30 B) 40 C) 45 D) 60

D4. A triangle has angles measuring 40° and 65° . What is the measure of the third angle?

- A) 65° B) 75° C) 85° D) 95°

D5. Two figures are similar with scale factor 3. A 5° angle in the smaller figure corresponds to an angle of what measure in the larger figure?

- A) 15° B) 5° C) 45° D) 3°

D6. (SPR) The legs of a right triangle have lengths 5 and 12. What is the length of the hypotenuse?

D7. An isosceles right triangle has a leg of length 5. What is the length of the hypotenuse?

- A) 5 B) $5\sqrt{2}$ C) $5\sqrt{3}$ D) 10

D8. In a 30-60-90 triangle the hypotenuse has length 14. What is the length of the shorter leg?

- A) 7 B) $7\sqrt{3}$ C) 14 D) $14\sqrt{3}$

D9. In a right triangle, the side opposite θ has length 3 and the hypotenuse has length 5. What is $\sin \theta$?

- A) $\frac{3}{4}$ B) $\frac{3}{5}$ C) $\frac{4}{5}$ D) $\frac{4}{3}$

D10. Which expression is equal to $\cos 20^\circ$?

- A) $\sin 20^\circ$ B) $\sin 70^\circ$ C) $\tan 20^\circ$ D) $\cos 70^\circ$

D11. (SPR) A circle has diameter 10. The circumference is $k\pi$. What is the value of k ?

D12. A circle has radius 6. What is the length of an arc with central angle 60° ?

- A) π B) 2π C) 6π D) 12π

D13. The graph of $(x - 2)^2 + (y + 1)^2 = 16$ in the xy -plane is a circle. What is the radius of the circle?

- A) 2 B) 4 C) 8 D) 16

D14. A tree 10 feet tall casts a 5-foot shadow. At the same time, a nearby tree casts a 2-foot shadow. What is the height, in feet, of the second tree?

A) 2 B) 4 C) 7 D) 25

D15. A rectangle has side lengths 8 and 6. What is the length of a diagonal?

A) 7 B) 10 C) 14 D) 48

7. Mixed module (SAT shape)

22 questions. Target **35 minutes**. Calculator / Desmos allowed. About the same mix as a real Math module: more Algebra and Advanced Math, fewer PSDA and Geometry, some SPR.

M1. $f(x) = 2x - 5$. What is $f(4)$?

A) 1 B) 3 C) 8 D) 13

M2. Which equation represents a line with slope $-\frac{1}{2}$ that passes through $(0, 6)$?

A) $y = 2x + 6$ B) $y = -\frac{1}{2}x + 6$ C) $y = -\frac{1}{2}x - 6$ D) $y = \frac{1}{2}x + 6$

M3. (SPR) A store charges \\$8 per notebook plus a \\$3 bag fee. The total is \\$35. How many notebooks were bought?

M4. The system $y = x + 1$ and $2x + y = 10$ has solution (x, y) . What is x ?

A) 3 B) 4 C) 5 D) 9

M5. How many solutions does $3x - 6y = 9$ and $x - 2y = 3$ have?

A) Zero B) One C) Two D) Infinitely many

M6. Which point is a solution of $y > 2x - 1$?

A) $(2, 2)$ B) $(1, 2)$ C) $(3, 4)$ D) $(0, -2)$

M7. Which expression is equivalent to $4x(x - 3) - 2(x - 3)$?

A) $2(x - 3)(2x - 1)$ B) $2(x - 3)(2x + 1)$ C) $4x^2 - 10x + 6$ D) $2(x - 3)(2x - 1) + 1$

M8. (SPR) What is the minimum value of $y = x^2 - 4x - 1$?

M9. The function h is defined by $h(t) = -16t^2 + 48t + 4$. What is the maximum value of $h(t)$?

A) 4 B) 16 C) 36 D) 40

M10. Which equation models a quantity that starts at 20 and decays by 15% each year?

A) $y = 20 - 15x$ B) $y = 20(0.15)^x$ C) $y = 20(0.85)^x$ D) $y = 20(1.15)^x$

M11. (SPR) What is the larger solution of $|2x - 3| = 7$?

M12. How many distinct real solutions does $\frac{x}{x-1} = \frac{2}{x-1} + 1$ have?

A) Zero B) One C) Two D) Infinitely many

M13. (SPR) What is the valid solution of $\sqrt{3x - 2} = 4$?

M14. In a group, 25% of the students play piano. Of those who play piano, 40% also play violin. What percent of all the students play both?

A) 10% B) 15% C) 65% D) 40%

M15. (SPR) A jacket costs \\$60. It is discounted 15%, then 8% tax is added to the **sale** price. What is the final price, in dollars? (Enter the exact decimal.)

M16. A poll is 61% with margin of error 5%. Which statement is **not** justified?

- A) A reasonable interval for the true percent is 56% to 66%.
- B) More than half of the population might favor the idea.
- C) Exactly 61% of the whole population favors the idea.
- D) A larger random sample would likely give a smaller margin of error.

M17. (SPR) Two printers run together at 40 and 55 pages per minute. How many minutes do they need to print 600 pages? Enter a decimal or a fraction.

M18. The angles of a triangle are x , $2x$, and $3x$. What is x , in degrees?

- A) 20 B) 30 C) 36 D) 60

M19. (SPR) A 30-60-90 triangle has short leg 6. What is the length of the hypotenuse?

M20. A right triangle has legs 9 and 12. What is the hypotenuse?

- A) 13 B) 15 C) 21 D) 108

M21. A circle has radius 8 and a central angle of 90° . What is the arc length?

- A) 2π B) 4π C) 8π D) 16π

M22. In the xy -plane, the circle $(x + 3)^2 + (y - 1)^2 = 9$ has center

- A) $(3, -1)$ B) $(-3, 1)$ C) $(3, 1)$ D) $(-3, -1)$
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8. Answer keys

Instructor side.

Algebra

A1. $f(2) = 9, g(2) = 14, 4 \cdot 9 - 14 = 22$. **C**

A2. $2y = -5x + 10, y = -\frac{5}{2}x + 5$. **A**

A3. 12 is dollars per extra shirt. **B**

A4. **B** $40 + 25m = 165$

A5. Add: $3x = 15, x = 5, y = 5$. **A**

A6. $a + c = 15, 12a + 7c = 145 \Rightarrow a = 8$

A7. Second $\times 2$ is $2x - 4y = 10$, not 6. Parallel, distinct. **A** (zero)

A8. $-2x > 8 \Rightarrow x < -4$. **B**

A9. Only $(1, 0.5)$ satisfies both. **C**

A10. $m = \frac{8}{4} = 2, y - 5 = 2(x - 2), y = 2x + 1$. **A**

A11. Perpendicular slope $-\frac{1}{3}$, through $(0, 4)$. **B**

A12. $2(5x - 2) = 4(x + 6) \Rightarrow x = \frac{14}{3}$

A13. $120 - 4t = 0 \Rightarrow t = 30$

A14. $6 + k > 10 \Rightarrow k > 4$. **A**

A15. $7 - 6x + 3 = 2x + 8 \Rightarrow x = \frac{1}{4}$. **A**

A16. $0.10t = 12 \Rightarrow t = 120$

Advanced Math

B1. Factor $x + 4$: $(x + 4)(2x - 1)$. **A**

B2. $\frac{(x - 3)(x + 3)}{x - 3} = x + 3$. **B**

B3. $9x^4y^2 \Rightarrow 9 + 4 + 2 = 15$

B4. Difference of cubes. **B**

B5. $x = 3$ or $x = -1$. **A**

B6. $g(x) = (x - 0)^2 + 55$. Minimum 55. **C**

B7. $x = -\frac{b}{2a} = 4$

B8. 9

B9. $x = 3$ or $x = -\frac{11}{3}$. **B**

B10. Absolute value cannot equal a negative. **A**

B11. $x + 3 = 2x - 2 \Rightarrow x = 5$

B12. Candidates 3 and -1 ; $\sqrt{1} \neq -1$. **B**

B13. $2^{x+1} = 2^4 \Rightarrow x = 3$. **B**

B14. $200 \times (1.5)^2 = 450$

B15. $x^2 - 2x - 3 = 0 \Rightarrow$ two points. **C**

B16. $3 \cdot 16 = 48$

PSDA

C1. 30

C2. 150

C3. $0.30 \times 0.40 = 0.12$. **B**

C4. $80 \times 1.25 = 100$, $100 \times 0.75 = 75$. **B**

C5. 40

C6. $800 \times 1.21 = 968$. **B**

C7. $72/1.2 = 60$

C8. Middle value 3. **B**

C9. Sum must be 400; known sum 320; fifth **80**

C10. Seniors only: 18/30. **B**

C11. **B**

C12. **B**

C13. $40/8 = 5$. **A**

C14. Over the grand total: 18/50. **B**

Geometry

D1. 27

D2. $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. **C**

D3. Area ratio 4 : 9, $20 \times \frac{9}{4} = 45$. **C**

D4. 75° . **B**

D5. Angles do not scale. **B**

D6. 13

D7. $5\sqrt{2}$. **B**

D8. Half the hypotenuse. **A**

D9. $\frac{3}{5}$. **B**

D10. $\cos \theta = \sin(90^\circ - \theta)$. **B**

D11. 10

D12. $\frac{1}{6} \cdot 12\pi = 2\pi$. **B**

D13. $r = 4$. **B**

D14. $\frac{10}{5} = \frac{h}{2} \Rightarrow h = 4$. **B**

D15. 10. **B**

Mixed module

M1. 3. **B**

M2. **B**

M3. $8n + 3 = 35 \Rightarrow n = 4$

M4. $2x + (x + 1) = 10 \Rightarrow x = 3$. **A**

M5. Second $\times 3$ is the first equation. Same line. **D**

M6. $2 > 1$ only for $(1, 2)$. **B**

M7. $(x - 3)(4x - 2) = 2(x - 3)(2x - 1)$. **A**

M8. Vertex at $x = 2, y = -5$

M9. $t = 1.5, h(1.5) = 40$. **D**

M10. Multiply by 0.85 each year. **C**

M11. $x = 5$ (the other is -2)

M12. Contradiction after clearing; $x = 1$ forbidden. **A**

M13. $3x - 2 = 16 \Rightarrow x = 6$

M14. $0.25 \times 0.40 = 0.10$. **A**

M15. $60 \times 0.85 = 51, 51 \times 1.08 = 55.08$

M16. **C**

M17. $600/95 = \frac{120}{19}$ (about 6.32)

M18. $6x = 180 \Rightarrow x = 30$. **B**

M19. 12

M20. 9-12-15. **B**

M21. $\frac{1}{4} \cdot 16\pi = 4\pi$. **B**

M22. Center $(-3, 1)$. **B**